JOSUE TORRES-FONSECA

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EDUCATION

Doctor of Philosophy in Computer Science and Engineering **Expected May 2028**
University of Michigan, Ann Arbor, MI **GPA: 4.0**

M.S. Computer Science and Engineering **Graduated May 2025**
University of Michigan, Ann Arbor, MI **GPA: 4.0**

B.S. Computer Science, Minor: Applied Mathematics **Graduated May 2023**
Boise State University, Boise, ID **GPA: 3.889; Magna Cum Laude**

QUALIFICATIONS

- Proficient in Python and Java; working knowledge of C#, SQL, JavaScript, React/Redux, .NET, and Git
- Machine learning with PyTorch, NumPy, Pandas, and Matplotlib
- Large language models and vision-language models: fine-tuning (QLoRA), knowledge distillation, evaluation, and on-premise serving with vLLM
- Embodied AI and robotics: AI2-THOR, PDDL planning, SLAM, and the Anki Cozmo and Misty II robots
- Designing and running user studies; Agile development, teamwork, and effective communication

WORK EXPERIENCE

University of Michigan **August 2023 – Present**
Graduate Researcher, Situated Language and Embodied Dialogue Group; supervised by Dr. Joyce Chai

- **SafetyALFRED (Findings of ACL 2026):** Created a safety-centric dataset using PDDL to generate trajectories that complete household tasks while mitigating six categories of kitchen hazards across 30 AI2-THOR scenes
- Evaluated 11 state-of-the-art VLMs on hazard recognition and safe embodied planning, finding models recognized hazards up to 92% of the time but fixed them less than 60% of the time
- Developed a decoupled multi-agent framework separating hazard recognition from task execution, raising the rate at which Qwen3-VL-32B fixed appliance-misuse hazards from 0.7% to 71.1%; open-sourced the code and dataset
- **ReMI (under review at HRI 2027, co-first author):** Designed and built an embodied conversational robot that encourages individuals living with cognitive decline to reflect on their resilience, co-designed with 10 dementia experts and 14 people with cognitive impairment
- Developed a structured dialogue manager whose pacing blinded raters scored above end-to-end LLM prompting in 35 of 40 simulated conversations; deployed in 7 lab and 3 in-home studies
- Fine-tuned Qwen3-VL-32B on 15K dialogue turns distilled from Gemini 2.5 Pro, plus a resilience topic classifier reaching 84% accuracy; served both on-premise with vLLM to protect participant privacy

Boise State University **August 2021 – August 2023**
Undergraduate Research Assistant, Speech Language and Interactive Machines Group; supervised by Dr. Casey Kennington

- **Object permanence (SIGdial 2022):** Built an incremental spoken dialogue system on the Anki Cozmo robot that transcribes and understands speech, detects and learns about objects, and maps them using SLAM
- Extended Cozmo's map to remember each object it had seen and where; in an A/B study with 24 participants, this raised the robot's accuracy answering questions about learned objects from 70.0% to 82.7%
- **HADREB (LREC 2022):** Collected and analyzed a dataset of over 1,000 robot emotional behaviors described and rated by 32 participants on the Misty II and Cozmo robots; showed a zero-shot language model predicts perceived emotion with up to 79.3% accuracy

Massachusetts Institute of Technology **May 2022 – November 2022**
Undergraduate Research Assistant, Language & Intelligence Group; supervised by Dr. Jacob Andreas

- Designed and built an interpretable machine learning model, extending the MILAN pipeline, that provides neuron annotations without training data

- Presented a poster and delivered a lightning talk at the end of the MIT Summer Research Program

Paylocity **May 2021 – August 2021**

Software Engineer Intern

- Designed, developed, tested, and deployed new features in project releases as part of a 13-member team
- Worked with product owners and leads to translate use cases into features
- Applied Agile and Scrum methodologies; gained full-stack experience with JavaScript, React/Redux, C#, .NET, SQL, and Git

Texas State University **June 2020 – August 2020**

Undergraduate Research Assistant, Intelligent Security Group; supervised by Dr. Mina Guirguis (NSF REU)

- Designed and built a game and reward system to secure a simulated water tank using game theory
- Implemented check blocks acting as both an intrusion detection and prevention system for the simulated tank

Boise State University **August 2020 – May 2023**

Tutor and Lab Assistant

- Tutored Data Structures (Fall 2020), Introduction to Theory of Computation (Spring 2021), and Operating Systems (Spring 2023); assisted with Microprocessors labs (Fall 2020)
- Led individual and group sessions, helped students plan and debug programming assignments and hardware labs, and connected course concepts to industry practice

PUBLICATIONS

- **SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models.** J. Torres-Fonseca, N. Deng, Y. Dai, S. Storks, Y. Zhang, R. Mihalcea, C. Kennington, J. Chai. *Findings of ACL 2026.*
- **The Mechanistic Emergence of Symbol Grounding in Language Models.** S. Wu, Z. Ma, X. Luo, Y. Huang, J. Torres-Fonseca, F. Shi, J. Chai. *ICML 2026.*
- **rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms.** A. Manaseryan, P. Rigby, B. Matthews, C. Henry, J. Torres-Fonseca, R. Whetten, E. Levandovsky, C. Kennington. *SIGdial 2025.*
- **Symbol and Communicative Grounding through Object Permanence with a Mobile Robot.** J. Torres-Fonseca, C. Henry, C. Kennington. *SIGdial 2022.*
- **HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors.** J. Torres-Fonseca, C. Kennington. *LREC 2022.*

ORGANIZATIONS

Treasurer, Computer Science and Engineering Graduate Students **August 2024 – May 2025**

- Managed organization funds and assisted in organizing events

President, AI Club **August 2024 – May 2025**

- Facilitated professional and technical development workshops on Python, machine learning, and data science; managed club funds

President, Competitive Cybersecurity and Cyber-Physical Club **August 2019 – May 2022**

- Competed in the CyberForce Competition at Idaho National Laboratory (46th of 103 teams, Fall 2019); presented on data scraping and reverse engineering with Ghidra and led Capture the Flag activities

HONORS & AWARDS

Gemini Academic Program Award	**($10,000)**	**Spring 2026 – Spring 2027**
NSF Graduate Research Fellowship	**($159,000)**	**Fall 2023 – Spring 2027**
Dean's List		**Fall 2018 – Spring 2023**
Boise State University Top Ten Scholar		**Spring 2023**
Barry M. Goldwater Scholarship	**($7,500)**	**Fall 2022 – Spring 2023**
Federal Employee Education & Assistance Fund Scholarship	**($1,000)**	**Fall 2018**